

# Improving Text Classification using Transformer-based Models in NLP

## Abstract

Natural Language Processing (NLP) enables machines to understand human language. This paper explores transformer-based models such as BERT for text classification.

## Introduction

NLP is a field of Artificial Intelligence focused on language understanding. It is used in translation, chatbots, and sentiment analysis.

## Methodology

We use a transformer model with tokenization, embeddings, fine-tuning and classification layers.

## Results

The model achieves high accuracy and outperforms classical machine learning methods.

## Conclusion

Transformer models significantly improve NLP performance and open new research directions.